

PRESERVING NATIVE LIPIDS AT SCALE: AUTOMATED COPOLYMER NANODISC PURIFICATION FOR MEMBRANE PROTEIN FUNCTION AND STRUCTURE

Anton STARKE and Lea ESSER

Cube Biotech, Creative Campus Monheim; Creative-Campus-Allee 12, 40789 Monheim, Germany

Anton.Starke@cube-biotech.com

Membrane proteins occupy a central role in modern life-science research because they define how cells sense, communicate with, and adapt to their environment. As receptors, channels, transporters, enzymes, and adhesion molecules, they regulate signal transduction, ion and metabolite flux, membrane excitability, intercellular communication, and lipid-dependent cellular homeostasis. Their relevance is also strongly reflected in drug discovery: more than half of current FDA-approved drug targets are integral membrane proteins, and GPCRs alone account for approximately 35% of approved drugs.

Despite their importance, membrane proteins remain challenging to purify in stable, functional, and structurally tractable forms. For decades, detergent-based workflows have been the standard approach for solubilization and purification. While detergents have enabled major progress, they replace the native lipid bilayer with micellar environments that can remove annular and regulatory lipids, weaken stabilizing protein-lipid interactions, and destabilize conformational states relevant for ligand binding, signaling, and assay development.

Amphiphilic copolymers provide a powerful alternative by extracting membrane proteins directly from biological membranes together with surrounding endogenous lipids, forming native nanodiscs that better preserve the physiological protein-lipid interface. However, because the optimal copolymer environment depends on target identity, membrane composition, polymer chemistry, and downstream application, copolymer selection has remained a labor-intensive empirical process.

Here, we present automated, plate-based workflows for copolymer screening and membrane protein purification that enables scalable parallel processing in a multi-well

format. These workflows reduce hands-on time, improve reproducibility, and facilitate rapid identification of suitable native nanodisc conditions. We demonstrate the applicability of this strategy across a diverse panel of membrane protein targets (GPCRs, solute carriers, ion channels, claudins, and cell-surface receptors), highlighting improved yields, particle homogeneity, and downstream compatibility compared with conventional approaches. Together, this work establishes a streamlined and scalable framework for studying membrane proteins in native-like lipid environments, from individual targets toward broader exploration of the membrane proteome.